

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech. (CST)(CBCS) Degree Examinations
 January 2023

22CST132 - DESIGN OF TALL STRUCTURES

Duration: 3 Hours

Max. Marks:100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of **IS:875, IS:1893 and IS:16700** is permitted.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Enumerate the different types of loads to be considered in the analysis and design of tall buildings.	10	L*2	1	1,2
	b) Write a short note on:				
	i. High performance concrete	10	L2	1	1,2
	ii. Light weight concrete				
2.	a) Discuss the Earthquake resistant design philosophy. Explain DBE and MCE.	10	L2	2	1,2
	b) Explain the step by step procedure for determining the lateral loads acting on a tall building due to an earthquake using equivalent static force method.	10	L2	2	1,2
3.	Determine the design equivalent static wind forces on a RCC multistorey commercial complex building 12 m x 18 m x 51 m, situated in Mumbai. It is proposed to be constructed about 200 m inside the sea front. Take average storey height as 3 m and frames are spaced at 6 m c/c in both the directions. The building is oriented with its smaller dimension facing the sea.	20	L4	2	1,2
Unit – II					
4.	Mention the different structural systems adopted for tall buildings. Outline the performance of the structural systems under lateral loading.	20	L4	3	1,2
5.	a) Discuss the modelling techniques of structures for approximate analysis.	10	L2	4	1,2
	b) Outline the procedure of design for differential movement.	10	L2	4	1,2
6.	Discuss in detail the effects of creep, shrinkage, temperature and fire on the behavior of tall buildings.	20	L2	4	1,2
Unit – III					
7.	a) Write a note on				
	i) Torsional instability	10	L2	5	1,2
	ii) Out of plumb effects				
	b) Discuss in detail the behavior of symmetric wall frames with the help of a case study.	10	L2	5	1,2
8.	a) Explain the second order effects of gravity loading.	10	L2	5	1,2
	b) Describe the methods of overall buckling analysis of frames.	10	L3	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
